Confluent Cloud Terraform Deployment — Row-Driven PTF UDFs (early access example)

> This example deploys **two row-driven PTF UDFs** (source in examples/ptf_udf_row_driven/java/) to **Confluent Cloud** using Terraform: `UserEventEnricher` (set semantics, stateful per-user session tracking) and `OrderLineExpander` (row semantics, stateless one-to-many expansion). Both UDFs ship in the same uber JAR and run as two independent long-running Flink statements in the same compute pool. All infrastructure (environment, Kafka cluster, Flink compute pool, service accounts, API keys) and Flink SQL statements are declared as Terraform resources.

Table of Contents

- **1.0 Overview**
  - **1.1 How this differs from the other deployment paths**
- **2.0 How it works**
  - **2.1 Infrastructure provisioning**
  - **2.2 JAR delivery**
  - **2.3 Statement flow**
- **3.0 Prerequisites**
- **4.0 How to run**
  - **4.1 Deploy**
  - **4.2 Monitor**
  - **4.3 Tear down**
- **5.0 Resources**

1.0 Overview

1.1 How this differs from the other deployment paths

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Where it runs	Confluent Cloud	Confluent Platform on Minikube
How SQL is submitted	`confluent_flink_statement` Terraform resources	`sql-client.sh -f` on the JobManager pod
UDF JAR delivery	`confluent_flink_artifact` (uploaded to CC)	`kubectl exec` to Flink pods
Entry point	`make deploy-cc-ptf-udf-row-driven`	`make deploy-cp-ptf-udf-row-driven`
Requires code compilation	✅ (Java + Gradle for UDF JAR)	✅ (Java + Gradle for UDF JAR)

Aspect	Confluent Cloud Terraform (this example)	Confluent Platform SQL Client
Statement lifecycle	Managed by Terraform state	Managed by Flink session cluster
Same codebase for both?	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)	✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment)

2.0 How it works

2.1 Infrastructure provisioning

Terraform declares the full Confluent Cloud stack:

Resource	Purpose
`confluent_environment`	Isolated CC environment (`ptf-udf`)
`confluent_kafka_cluster`	Standard single-zone Kafka cluster (AWS us-east-1)
`confluent_schema_registry_cluster`	Stream Governance Essentials (auto-provisioned with environment)
`confluent_service_account`	`ptf-udf` service account with FlinkDeveloper, ResourceOwner, and Assigner roles
`confluent_flink_compute_pool`	Flink compute pool (max 10 CFU)
`confluent_api_key`	Flink-specific API key for statement submission
`flink_api_key_rotation` module	Rotating API key pair for the Flink service account

2.2 JAR delivery

On Confluent Cloud, UDF JARs are uploaded as **Flink artifacts** via the `confluent_flink_artifact` resource. The JAR is built from examples/ptf_udf_row_driven/java/ and uploaded directly from the local build output. The `CREATE FUNCTION ... USING JAR` statement references the artifact using a `confluent-artifact://` URI.

2.3 Statement flow

Terraform manages the SQL statements for **both** PTF UDFs (`UserEventEnricher` and `OrderLineExpander`) as `confluent_flink_statement` resources with explicit `depends_on` ordering. Both UDFs are bundled in the **same uber JAR**, so a single `confluent_flink_artifact` upload (Step 11) feeds both `CREATE FUNCTION` statements.

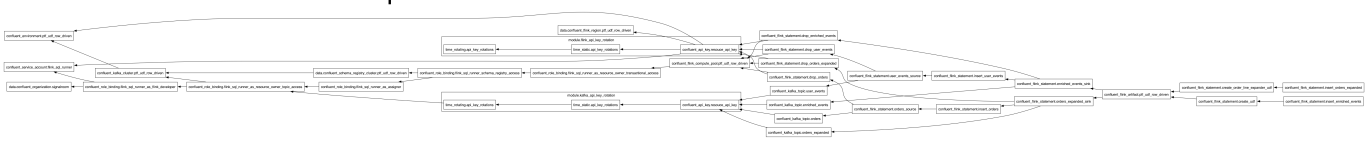
```
┌──────────────────────────────────────────────────────────────────────────────
│  ── UDF 1: UserEventEnricher (set semantics) ──
```

```
│
│  Step  1:  DROP TABLE IF EXISTS user_events                  → OK
│
│  Step  2:  CREATE TABLE user_events (...)                    → OK
│
│  Step  3:  INSERT INTO user_events VALUES (sample data)      → submitted
│
│  Step  4:  DROP TABLE IF EXISTS enriched_events              → OK
│
│  Step  5:  CREATE TABLE enriched_events (...)                → OK
│
│
│  ── UDF 2: OrderLineExpander (row semantics) ──
│
│  Step  6:  DROP TABLE IF EXISTS orders                       → OK
│
│  Step  7:  CREATE TABLE orders (...)                         → OK
│
│  Step  8:  INSERT INTO orders VALUES (sample data)           → submitted
│
│  Step  9:  DROP TABLE IF EXISTS orders_expanded              → OK
│
│  Step 10:  CREATE TABLE orders_expanded (...)                → OK
│
│
│  ── Shared artifact + both pipelines ──
│
│  Step 11:  Upload UDF JAR as confluent_flink_artifact        → OK
│
│            (contains BOTH ptf.UserEventEnricher and
│
│             ptf.OrderLineExpander)
│
│  Step 12:  CREATE FUNCTION user_event_enricher               → OK
│
│            AS 'ptf.UserEventEnricher'
│
│            USING JAR 'confluent-artifact://...'
│
│  Step 13:  CREATE FUNCTION order_line_expander               → OK
│
│            AS 'ptf.OrderLineExpander'
│
│            USING JAR 'confluent-artifact://...'
│
│  Step 14:  INSERT INTO enriched_events                       → submitted
│
│            SELECT ... FROM TABLE(
│
│              user_event_enricher(
│
│                input => TABLE user_events PARTITION BY user_id
```

```
│                )
│
│             )
│
│   Step 15:  INSERT INTO orders_expanded                      → submitted
│
│             SELECT ... FROM TABLE(
│
│               order_line_expander(
│
│                 input => TABLE orders
│
│               )
│
│             )         -- note: NO PARTITION BY (row semantics)
│
└──────────────────────────────────────────────────────────────────────────
┘
```

Steps 14 and 15 are **two long-running streaming jobs**, both running continuously side-by-side from the same Flink compute pool: Step 14 reads from `user_events` and writes per-user enriched output to `enriched_events`; Step 15 reads from `orders` and writes one expanded line-item row per order item to `orders_expanded`.

Once deployment completes, Terraform generates a visual **resource graph** at `examples/ptf_udf_row_driven/cc_deploy/terraform.png`, providing an at-a-glance view of the infrastructure and resource dependencies:



3.0 Prerequisites

- macOS with Homebrew or Linux with apt-get
- Java 21 and Gradle installed (for building the UDF JAR)
- Terraform installed
- A Confluent Cloud account with a **Cloud API key** and **secret** (create one here)
- The UDF JAR must be built before deploying:

```
make build-ptf-udf-row-driven   # builds the UDF uber JAR from
examples/ptf_udf_row_driven/java/
```

4.0 How to run

All commands are run from the **project root** (where the `Makefile` lives).

4.1 Deploy

A single target builds the UDF JAR and runs `terraform apply`:

```
make deploy-cc-ptf-udf-row-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

Behind the scenes this runs:

Step	What it does
1	`./gradlew clean shadowJar` — builds the UDF uber JAR from `examples/ptf_udf_row_driven/java/`
2	`terraform init` — initializes the Terraform working directory
3	`terraform apply -auto-approve` — provisions all CC infrastructure and submits Flink SQL statements
4	Generates a Terraform visualization at `docs/images/terraform-visualization.png`

4.2 Monitor

Monitor the running Flink statements in the Confluent Cloud Console:

1. Navigate to your **ptf-udf** environment
2. Open the **Flink** tab to see the compute pool and **two long-running statements** (the `INSERT INTO enriched_events ...` and the `INSERT INTO orders_expanded ...`)
3. Open the **Topics** tab to inspect all four topics: `user_events`, `enriched_events`, `orders`, and `orders_expanded`

4.3 Tear down

To destroy all Confluent Cloud resources created by Terraform:

```
make teardown-cc-ptf-udf-row-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

This runs `terraform destroy -auto-approve`, removing all Flink statements, the compute pool, Kafka cluster, service accounts, and the environment.

5.0 Resources

- Confluent Terraform Provider
- confluent_flink_statement Resource
- confluent_flink_artifact Resource
- Create a User-Defined Function with Confluent Cloud for Apache Flink